Problem Solving

1. Let $X_1, \ldots, X_n$ be a random sample from each of the distributions having the following pdfs or pmfs:

(a) $f(x;\theta) = \theta^x e^{-\theta}/x!$, $x = 0, 1, 2, \ldots$, $0 \leq \theta < \infty$, zero elsewhere.

(b) $f(x;\theta) = \theta x^{\theta-1}$, $0 < x < 1$, $0 < \theta < \infty$, zero elsewhere.

(c) $f(x;\theta) = e^{-(x-\theta)}$, $\theta \leq x < \infty$, $-\infty < \theta < \infty$, zero elsewhere.

In each case find the MLE $\hat{\theta}$ of θ.

2. Let the table represent a summary of a sample of size 50 from a binomial distribution having $n = 5$. Find the MLE of $P(X > 3)$.

x	0	1	2	3	4	5
Frequency	6	10	14	13	6	1

3. Let the table represent a summary from a sample of size 50 from a Poisson distribution. Find the MLE of $P(X = 2)$.

x	0	1	2	3	4	5
Frequency	7	14	12	13	6	3

4. Let $X_1, \ldots, X_n$ be a random sample from a distribution with one of two pdfs. If $\theta = 1$, then $f(x; \theta = 1) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, $-\infty < x < \infty$. If $\theta = 2$, then $f(x; \theta = 2) = 1/[\pi(1 + x^2)]$, $-\infty < x < \infty$. Find the MLE of θ.